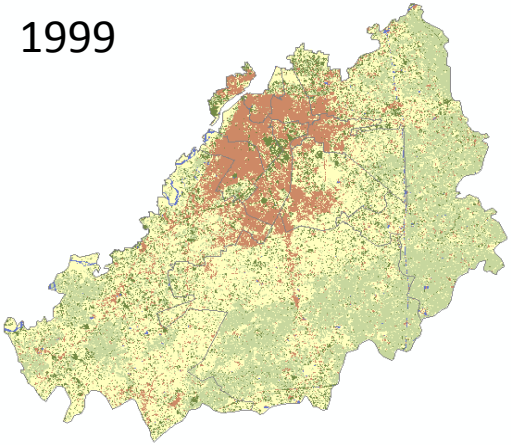
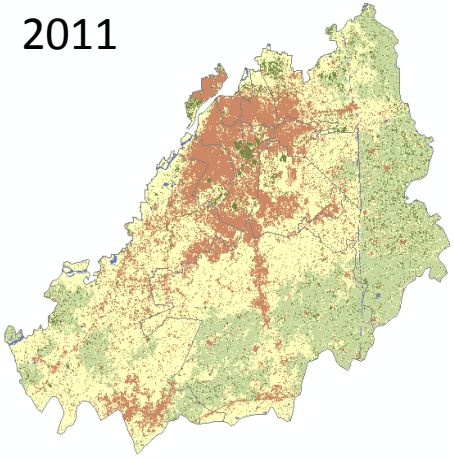
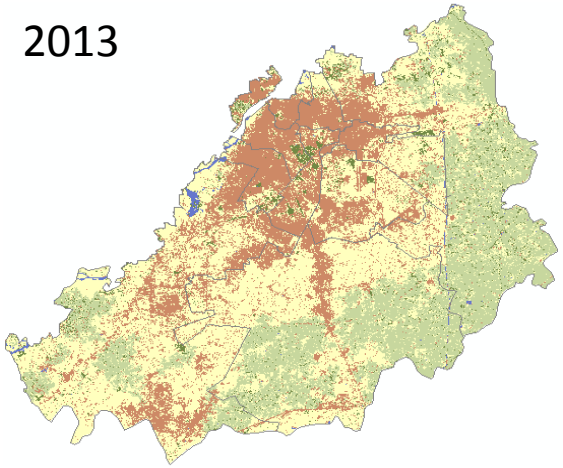
1999



2011

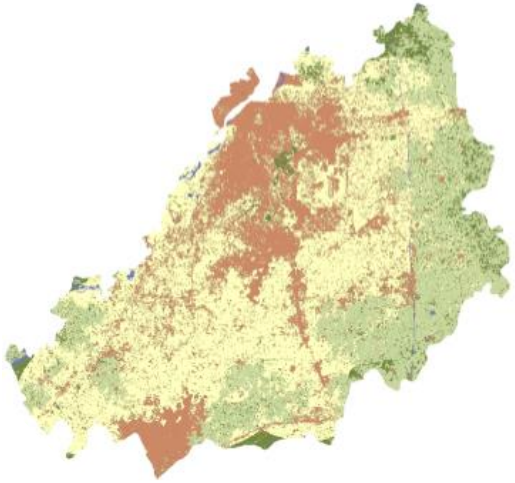


2013

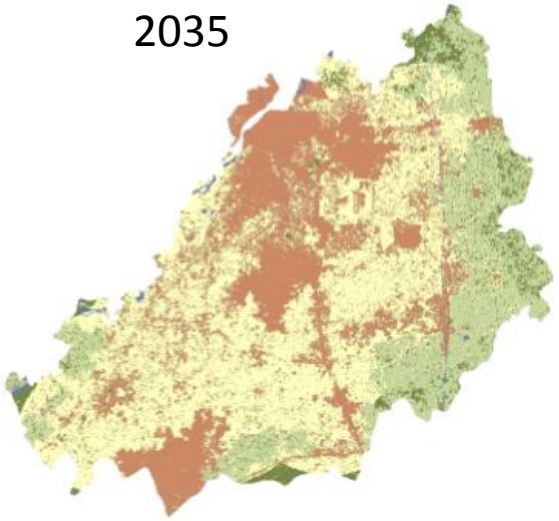


LAND USE MODELS-II

2021

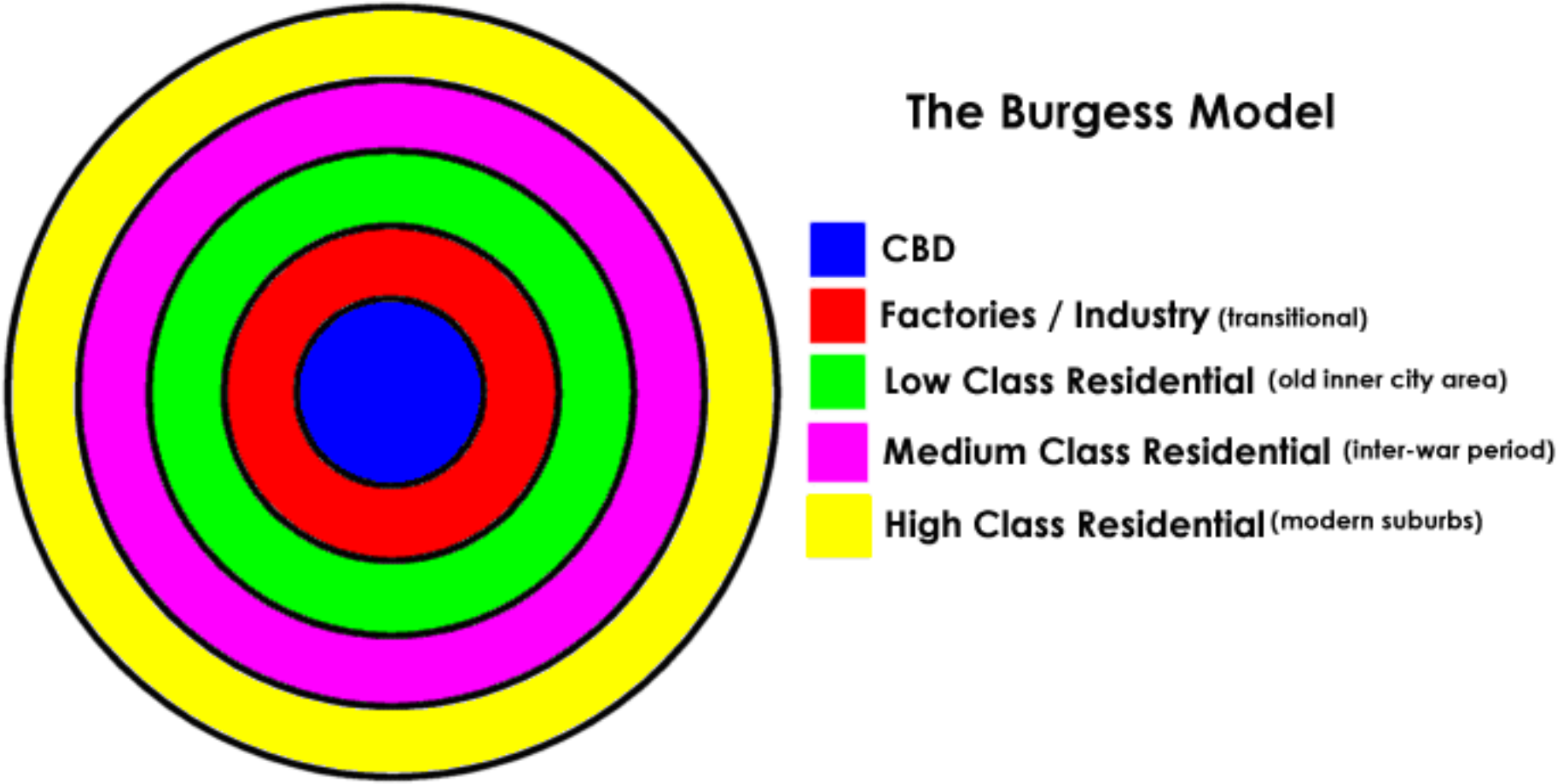


2035



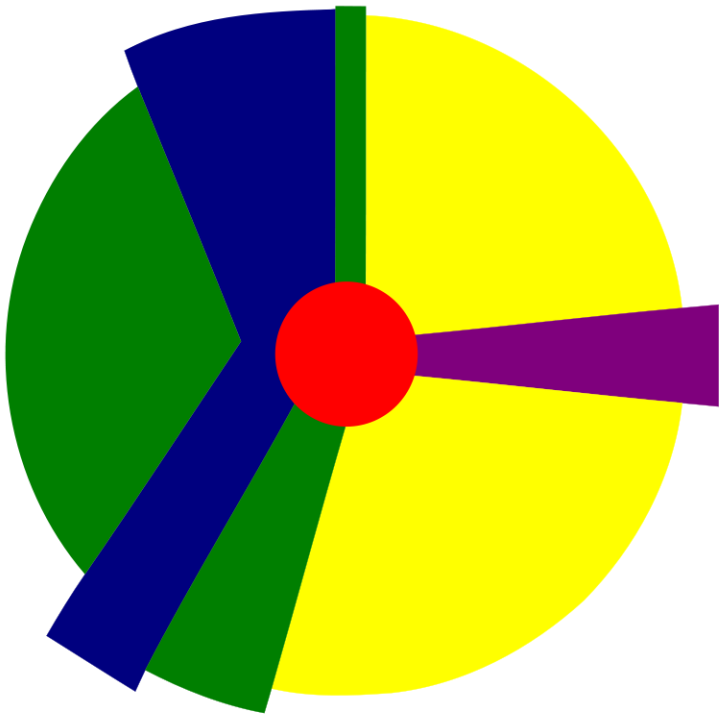
The Burgess Model

Ernest Burgess (1923)



The Sector Model

Homer Hoyt (1939)



Hoyt Sector Model Key

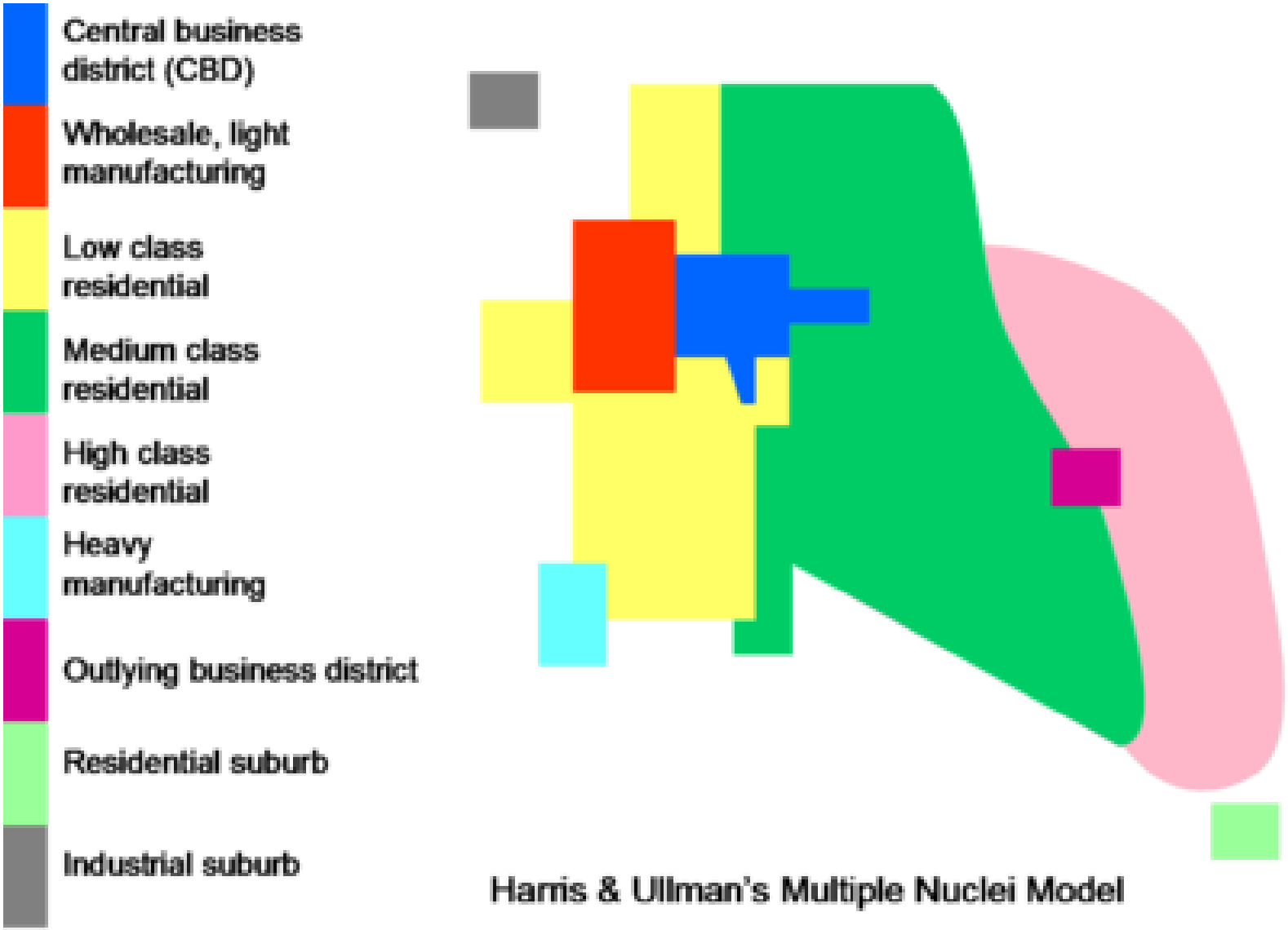
- CBD
- Factories/Industry
- Low class residential
- Middle class residential
- High class residential

Multiple-nuclei model

Main Emphasis

- **decentralization**
- **keep conflicting or competing land uses apart (e.g. high cost housing and smoky factories)**

Chauncy Harris and Edward Ullman, (1945) In *Nature of Cities*



Harris & Ullman's Multiple Nuclei Model

Multiple-nuclei model

- Distinctive land-use zones develop because some activities repel/ attract each other i.e.
  - high-quality housing does not generally arise next to industrial areas

- Large cities develop by the spread from several nodes, not just the one CBD.

- Individual nodes with specific functions (industrial, commercial, residential) developed originally due to benefits from clustering next to similar activities

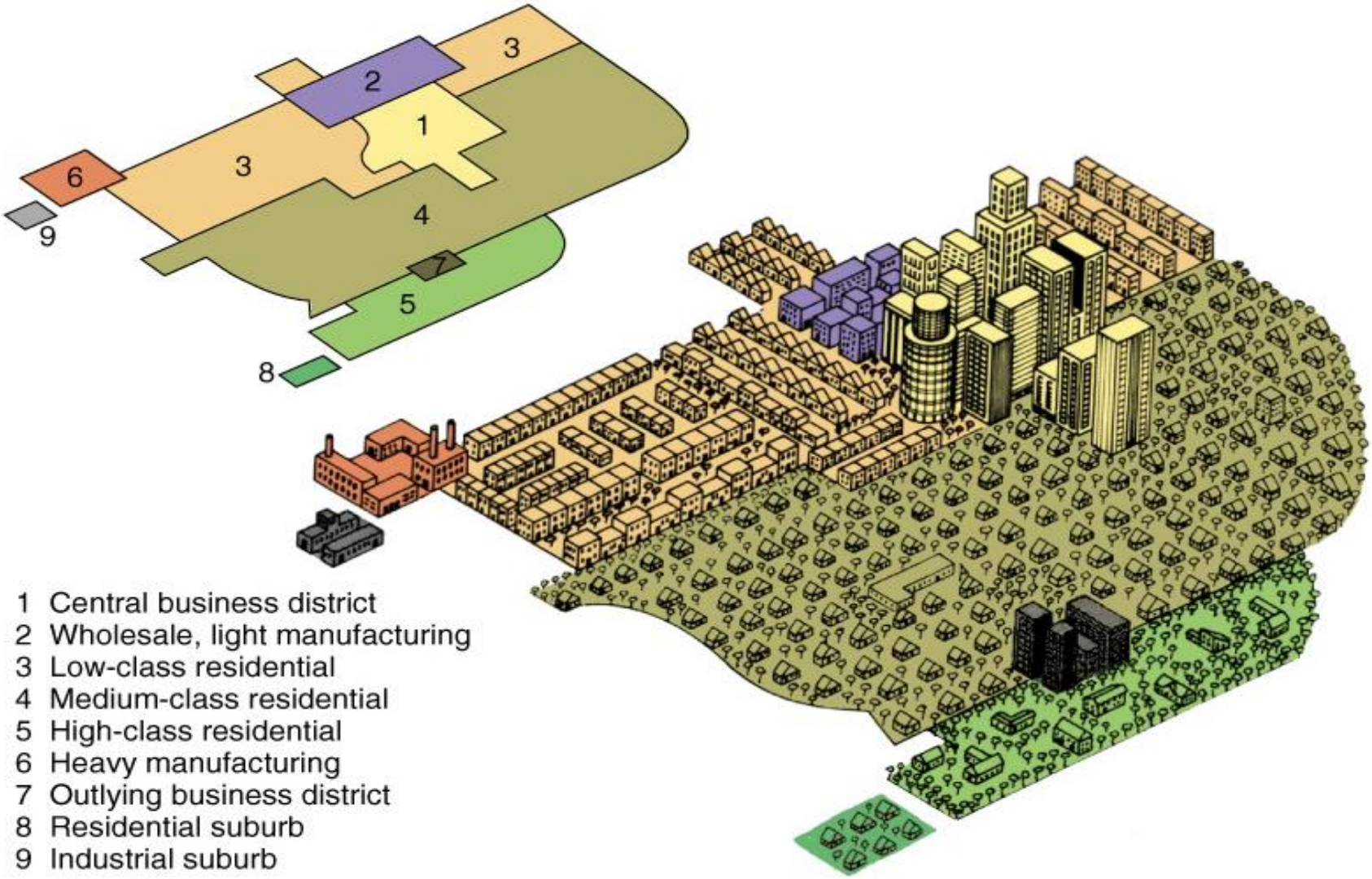
Multiple nuclei model

- The model has four geographic principles
  1) Certain activities require highly specialized facilities
     I. Accessible transportation for a factory
     II. Large areas of open land for a housing region
  2) Certain activities cluster because they profit from mutual association
  3) Certain activities repel each other and will not be found in the same area
  4) Certain activities could not make a profit if they paid the high rent of the most desirable locations

Assumptions

- The central business district will often be found near the original retail area of the city.
- Warehousing and light industrial areas will develop adjacent to the CBD and along transport links such as railways and roads.
- Heavy industry will locate on the outskirts of towns or where the outskirts
- Residential areas will be found in the remaining space.

Multiple nuclei model

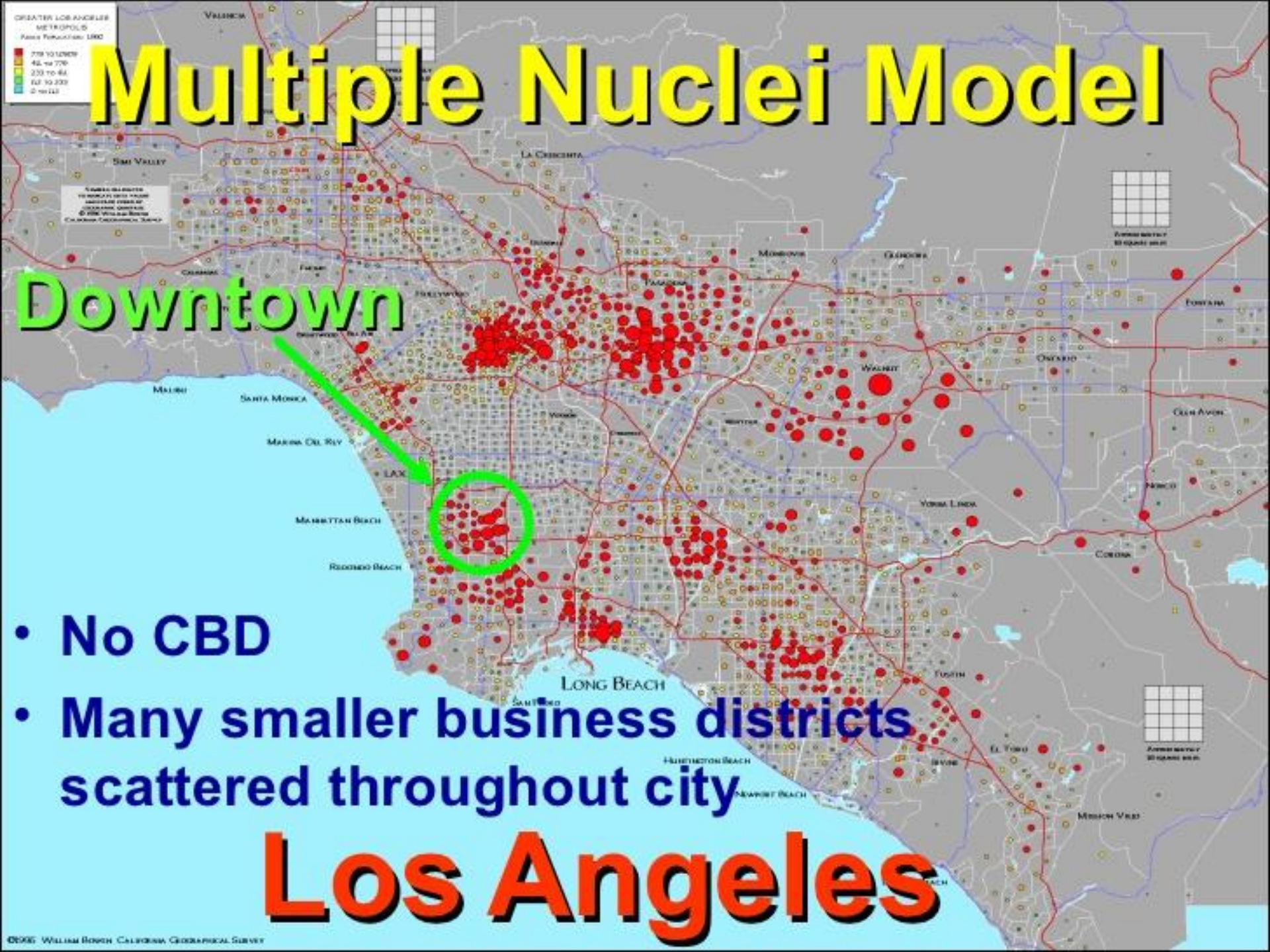


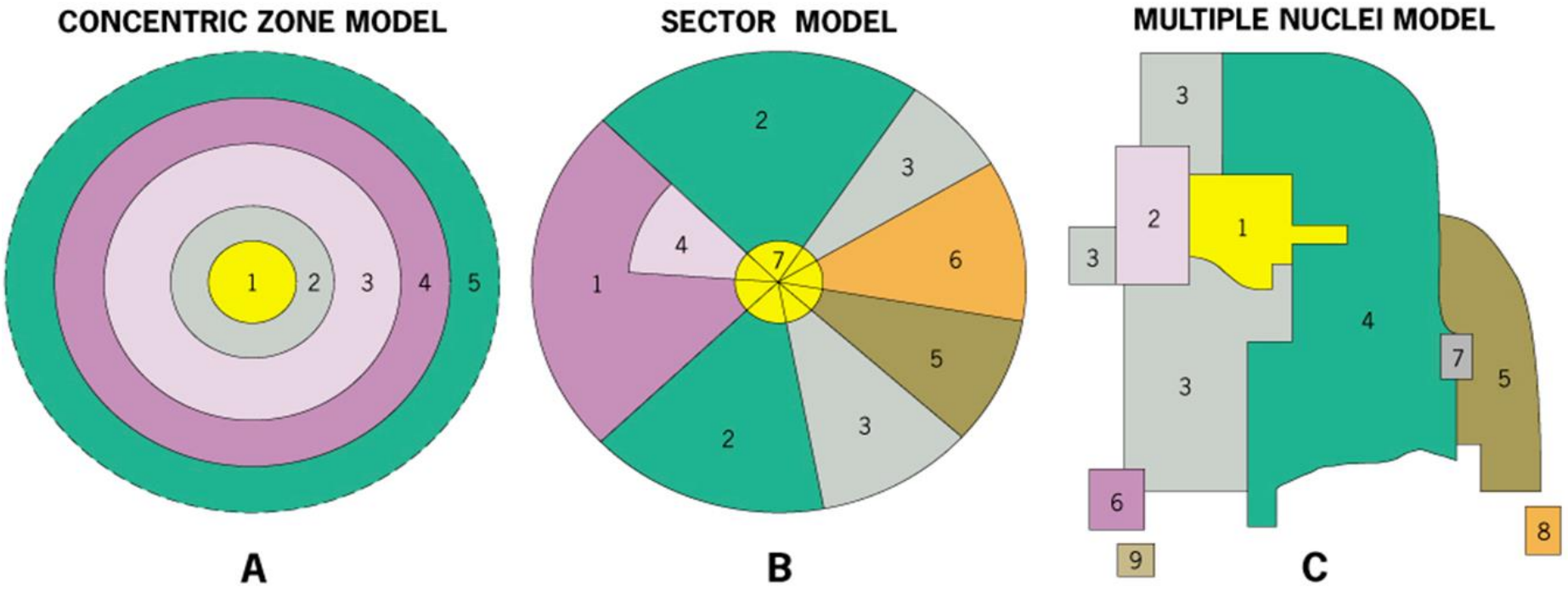
Stresses the importance of multiple nodes of activity, not a single CBD. Ports, airports, universities attract certain uses while repelling others.

Multiple nuclei model

Criticisms

- Each zone displays a significant degree of internal heterogeneity and not homogeneity
- No consideration of influence of physical relief and government policy.
- Not applicable to oriental cities with different cultural, economic and political backgrounds


Multiple Nuclei Model
Downtown
• No CBD
• Many smaller business districts scattered throughout city
Los Angeles


CONCENTRIC ZONE MODEL
1
2
3
4
5
A
SECTOR MODEL
1
2
3
4
5
6
7
B
MULTIPLE NUCLEI MODEL
1
2
3
4
5
6
7
8
9
C
1 Central business district
2 Zone of transition
3 Zone of independent workers' homes
4 Zone of better residences
5 Commuters' zone
1 High-rent residential
2 Intermediate-rent residential
3 Low-rent residential
4 Education and recreation
5 Transportation
6 Industrial
7 Core
1 Central business district
2 Wholesale, light manufacturing
3 Low-class residential
4 Middle-class residential
5 High-class residential
6 Heavy manufacturing
7 Outlying business district
8 Residential suburb
9 Industrial suburb

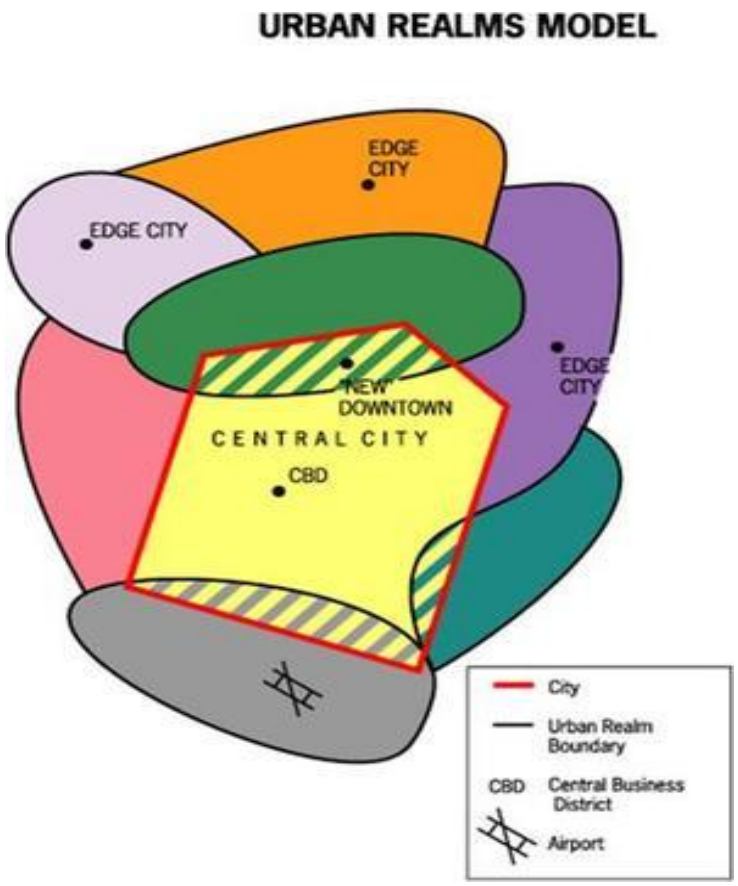
Urban Realms Model

Francisco Bay area

Developed by James E. Vance Jr. in the 1960's

- Each realm is a separate economic, social and political entity that is linked together to form a larger metro framework
- Suburbs are within the sphere of influence of the central city and its metropolitan CBD
- Now urban realms have become, so large they even have exurbs, not just suburbs

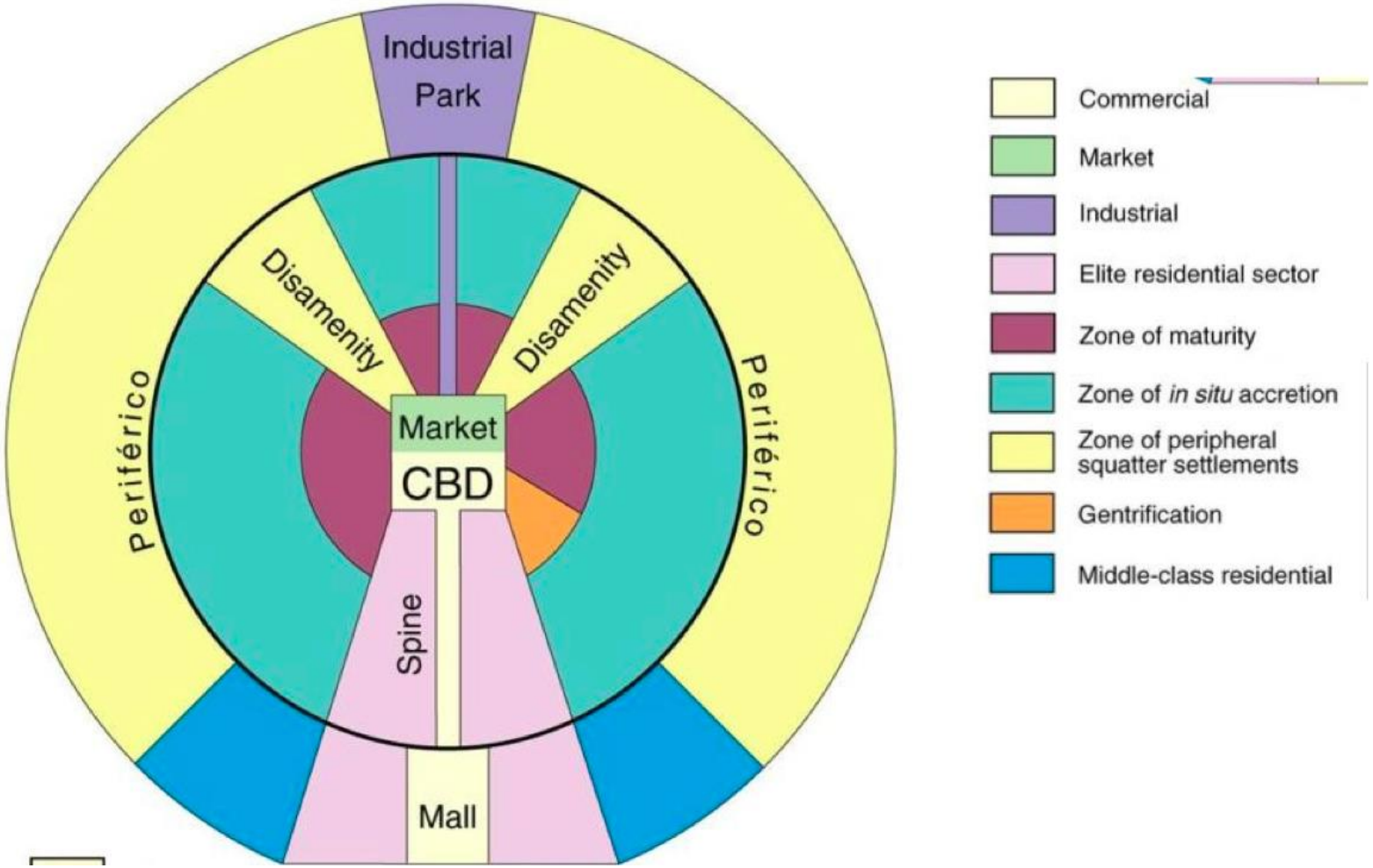
Urban realm depends on



- Overall size of the metropolitan region
- Amount of economic activity in each urban realm
- Topography and major land features
- Internal accessibility of each realm

Latin/South American City Models

Ernest Griffin and Larry Ford (1980/1996)

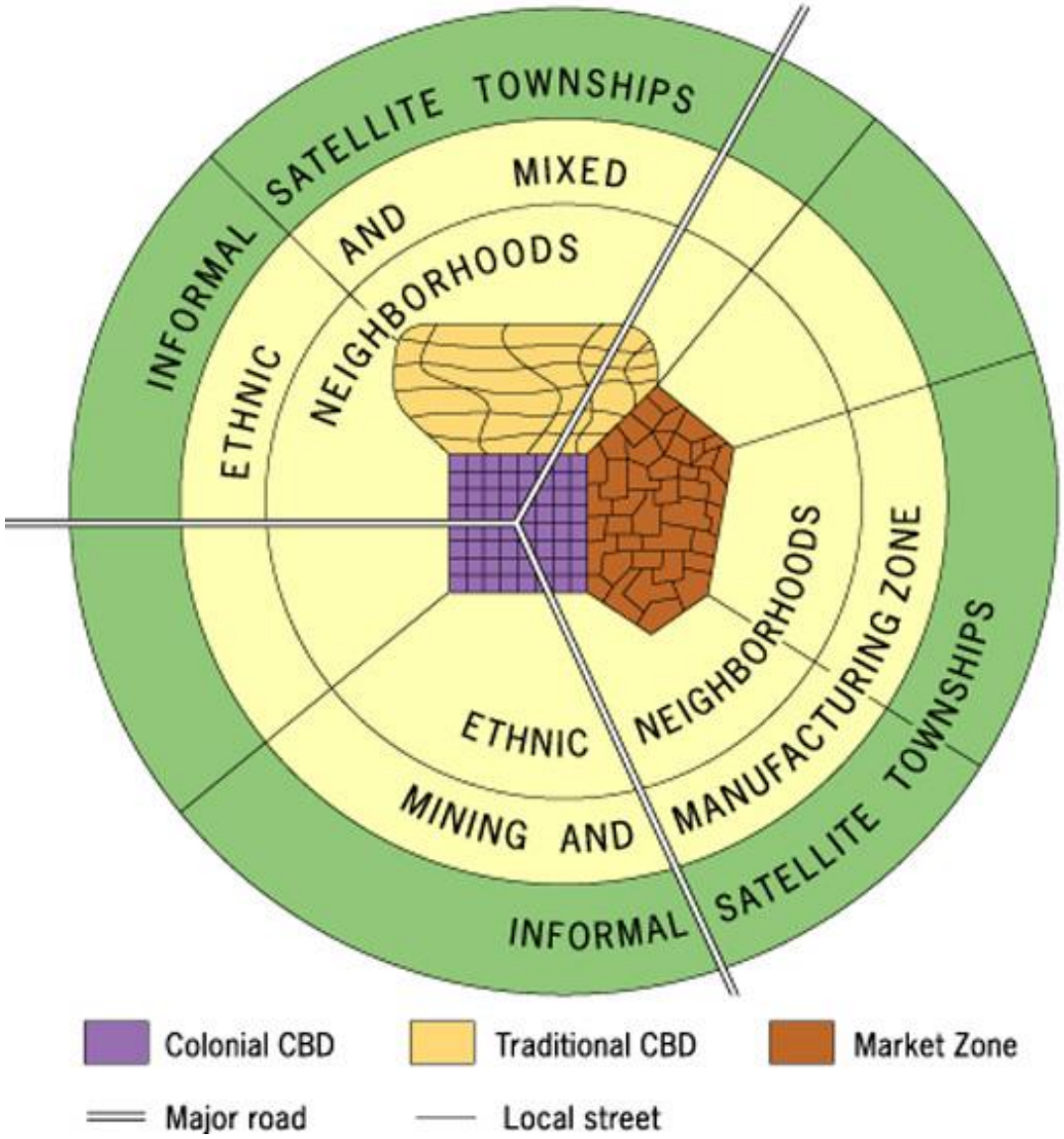


Ibero-American Model

- Latin American cities are growing rapidly-1950= 41% urban, 1997 = 74% urban
- CBD dominates the center with 2 main divisions-traditional market and modern high rises
- A commercial spine and axis of business is surrounded by elite residential housing
- The **spine** is an extension of the **CBD** with offices, shops, high class housing, restaurants, theaters, & parks
- **Zone of Maturity**-Middle class housing
- **Zone of In Situ Accretion** high pop. Density of modest housing
- **Periphery-Periferico**-high density shanty towns of extreme poverty and no services

Sub-Saharan African City Models

De Blij



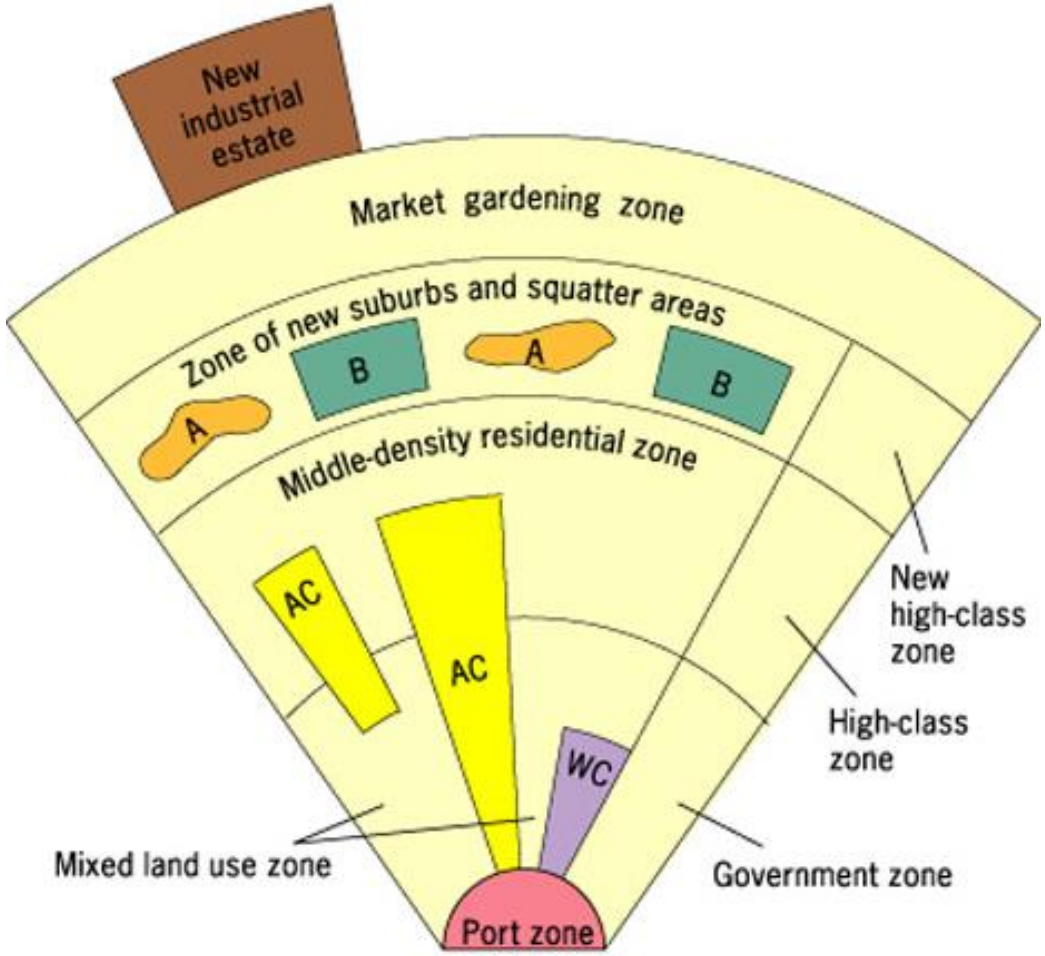
The African City

- African cities often have 3 CBDs=Colonial, Traditional and Periodic Market Zone
- Sub-Saharan Africa is the least urbanized area of the world, but the most rapidly urbanizing
- No large cities to match Cairo-Kinshasa, Nairobi, Harare, Dakar, Abidjan were established by Europeans
- South African cities-Johannesburg, Cape Town & Durbin are western cities with elements of European and American models-high rise CBDs and sprawling suburbs

Asian Models (South East Asia)

T.G. McGee

(1967)



The Southeast Asian City

- SE Asia-rapid growth of population & cities-1950-15% urban, 1990s-29% urban
- Most growth in coastal cities like Ho Chi Minh City (Saigon)
- Old colonial port zone surrounds the commercial district
- Unlike Western cities-no formal business zone, but separate clusters
- Kuala Lampur, Malaysia; Jakarta, Indonesia

Is there any Hope?

Propose your OWN IDEA


THANK
YOU!